

A Greener and More Scalable Synthesis of Biogenic Polydisulfides from Lipoic Acid

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Ring-opening polymerization (ROP) of 1,2-dithiolanes form polydisulfides, an emergent class of dynamic covalent polymers. However, both monomer and polymer syntheses typically require anaerobic and moisture-free conditions, often employing hazardous reagents and solvents that limit scalability. Herein, efficient, scalable syntheses for poly(ethyl lipoate) and ethyl lipoate that incorporate Green Chemistry principles are disclosed. The synthesis of ethyl lipoate from lipoic acid on a 100-gram scale (>80% yield) is optimized lowering the E-factor (2.27) by an order of magnitude compared to conventional methods. Diphenyl phosphate, a nonhazardous commercial organic acid, is used to synthesize ultra-high-molecular-weight poly(ethyl lipoate) on a 50-gram scale from cationic ROP (CROP). The polymerizations

proceed under ambient atmosphere in low-hazard and renewable solvents, and a mild depolymerization strategy to regenerate the monomer is developed. Due to their extreme molar mass, the materials possess unique mechanical and physical properties. Life cycle assessment (LCA) conducted on synthetic and recycling processes shows that the polydisulfide has competitive environmental impacts comparable to several commodity polymers, despite the latter having an efficiency advantage due to economies of scale. These discoveries establish an economical and scalable closed-loop polymer platform that can be broadly applied to various polydisulfides sourced from 1,2-dithiolanes such as lipoic acid.

1. Introduction

Research interest in creating a more sustainable and circular polymer economy is surging due to growing environmental impacts of legacy materials.^[1,2] Many complementary solutions are being pursued such as incorporating bio-derived feedstock^[3–5] and advanced recycling^[6–8] or upcycling approaches for handling polymer waste.^[9,10] Other recent advances in green polymer synthesis include using benign solvents,^[11–14] conducting polymerizations in bulk (i.e., without solvent),^[15] and adapting Click

chemistries^[16,17] for mild step-growth polymerizations. Despite these noteworthy advances, most examples fail to *holistically* incorporate Green Chemistry principles^[18,19] into polymer life-cycle design, yet this approach is vital to establishing a more sustainable polymer economy.^[20] However, a few noteworthy reports show a more comprehensive approach to creating new-to-the-world materials.^[21–24]

Greener polymers that can undergo chemical recycling to monomer (CRM) have the potential to create a closed-loop polymer economy with minimal waste and energy demands.^[25] Many polymer classes undergo CRM such as poly(thio)esters,^[26–29] polyalkenamers,^[30,31] and more recently polycarbonates^[32] or polyacetals.^[33] However, these polymers are generally derived from nonrenewable sources, may require intricate syntheses that do not adhere to Green Chemistry principles, and are often not scalable. On the other hand, polydisulfides formed from the ring-opening polymerization (ROP) of 1,2-dithiolanes^[34] undergo facile CRM and typically employ lipoic acid as an inexpensive biogenic feedstock.^[35–44] Moreover, copolymerization of lipoic acid and its derivatives with conventional acrylic monomers yields degradable vinyl-type polymers.^[45–49] These prior works reveal the versatility of 1,2-dithiolane monomers; however, radical and anionic ROP methods often employ Schlenk conditions and struggle to yield high-molecular-weight polymers (typically <80 kDa).^[37,50]

The cationic ROP (CROP) of cyclic disulfides has been known for decades,^[51,52] and recent work shows that this approach affords recyclable composites^[50] or adhesives^[53] and ultra-high-molecular-weight (UHMW) polymers^[40] in the MDa range. For the latter study, Qu and co-workers achieved UHMW polydisulfides by pairing organic acid catalysts such as trifluoroacetic acid (TFA) with a non-coordinating counterion co-catalyst. This

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approach required a multistep synthesis and polymerizations were conducted in chlorinated solvents under Schlenk conditions. Further, most methods for lipoate monomer synthesis employ Steglich-type couplings, which requires hazardous reagents and solvents limiting the scalability of monomer production. Nevertheless, the positive correlation between polymer molecular weight and robust thermomechanical properties^[54] make polydisulfides a valuable target for greener methods.

Herein, we describe a scalable, more sustainable process for a model polydisulfide employing Green Chemistry principles throughout the entire material life cycle. We resolve previous limitations in lipoate monomer synthesis and CROP protocols by developing a highly efficient Fischer esterification to yield ethyl lipoate (EtLp) at 100 gram-scale with minimal waste and high yield. We employ diphenyl phosphate (DPP) as a benign catalyst for the mild CROP of EtLp to afford UHMW poly(ethyl lipoate) (polyEtLp) under ambient atmosphere in "wet" acetone on the benchtop. Characterizing polymerization thermodynamics at different temperatures and monomer concentrations reveals optimal reaction conditions to synthesize polyEtLp on a 50-gram scale and subsequently recycle the polymer via ring-closing depolymerization (RCD). Laboratory-scale modeling via life cycle assessment (LCA), across several environmental impact metrics, shows that the production of polyEtLp may be competitive with several commodity polymers.

further they employed stoichiometric quantities of carbodiimides and organobases (e.g., 4-dimethylaminopyridine), which are both highly hazardous.^[55] In fact, DCM was recently reassessed by the U.S. Environmental Protection Agency (EPA), which now recommends lower thresholds for exposure limits and restrictions for its use in most settings.^[56] In our lab, we performed a Steglich esterification to afford EtLp in ~80% yield (10 gram-scale) with an E-factor >20; this is greater than values for typical commercial bulk-chemical process.^[57]

We then assessed Fischer esterification of lipoic acid with ethanol as an alternative method to mitigate waste, reduce hazards, and provide a more scalable process (100 gram-scale) for lipoate monomer production (Figure S63, Supporting Information). Although this synthetic method employs a catalytic amount of a strong acid, which can inadvertently polymerize the lipoic acid or monomer (*vide infra*), this can be mitigated by careful control of the reaction conditions. Quenching the reaction at elevated temperature, where the monomer is favored at equilibrium, with solid NaHCO₃ afforded spectroscopically pure EtLp with yields reaching up to 90% and E-factors as low as 2.27 (Figure S1, S2, Equation S1, Supporting Information). This E-factor value resembles those observed for industrial bulk chemical processes, which have the privilege of operating at much larger scales.^[57] Thus, by increasing the reaction scale, one can anticipate that the value of the E-factor for the Fischer esterification could be further decreased (Figure 1).

2. Results and Discussion

2.1. Green Monomer Synthesis

We chose EtLp as a model, benchmark monomer due to its history of synthesis by Steglich coupling methods.^[37,38] These protocols used dichloromethane (DCM) as the standard solvent;

2.2. Screening of Catalyst Activity for CROP

Several Brønsted acids (5 mol%) were initially screened for the CROP of EtLp under an ambient atmosphere on the benchtop in DCM at [EtLp]₀ = 4 M (Figure 2a; proposed mechanism shown in Scheme S1, Supporting Information). The reaction temperature at 30 °C ensured thermal stability throughout polymerization.

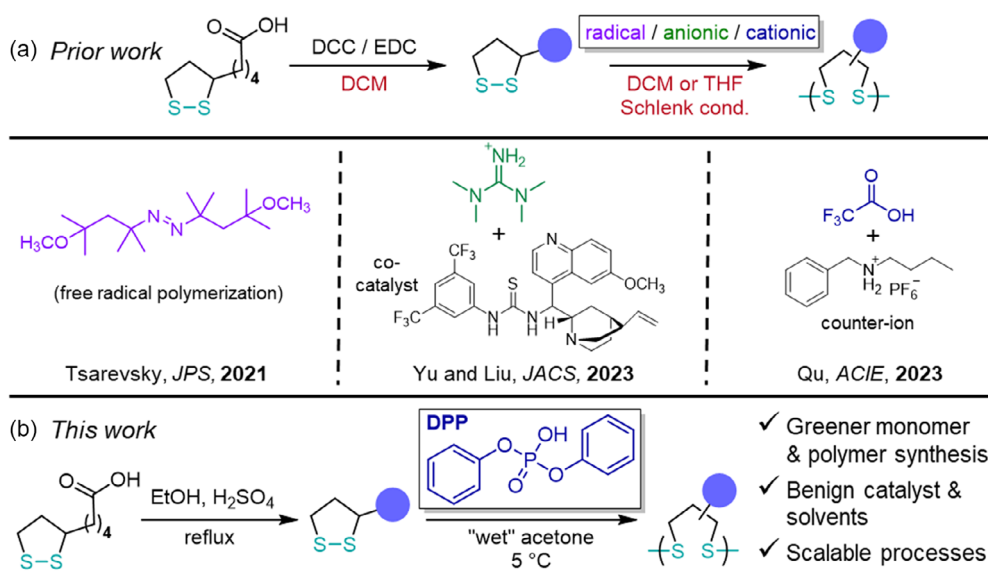


Figure 1. Synthesis and ROP of 1,2-dithiolanes derived from lipoic acid a) Typical methods used to synthesize lipoates and representative ROP methods. b) Overview of this work showing sustainability advantages to both monomer and polymer synthesis.

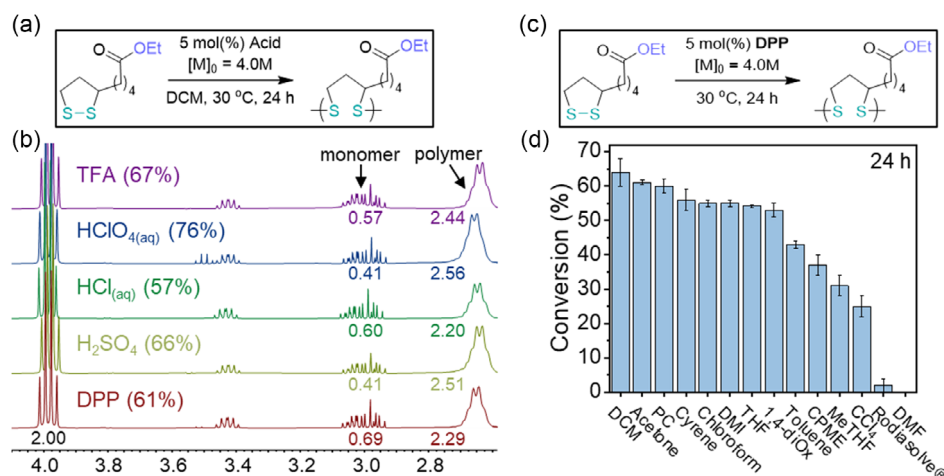


Figure 2. Screening catalyst and solvents for the CROP of EtLp. a) Reaction scheme for CROP of EtLp with various acids in DCM. b) ¹H NMR spectra of crude polyEtLp sampled at 24 h for various acid catalysts. c) Reaction scheme for CROP of EtLp with DPP in various solvents. d) Bar chart showing the monomer conversion in different solvents after 24 h. Standard deviation was calculated from $n = 3$ unique reactions.

Monomer conversion was monitored with ¹H NMR spectroscopy by assessing the relative molar ratio of formed polymer ($\delta = 2.65$) to remaining monomer ($\delta = 3.0$) (Table S1, Supporting Information). All reactions reached >50% conversion at 24 h, indicating tolerance of ambient water and oxygen (Figure 2b). We also observed no detectable hydrolysis of the ethyl ester functionality in EtLp to form lipoic acid during the polymerization. As expected, the polymers possessed regiorregularity^[39] in their backbone from competing H-H, H-T, and T-T monomer additions as evidenced by ¹³C NMR spectroscopy (for example, see Figure S48, Supporting Information).

The acceptable catalytic activity of all screened acids, despite their large differences in pK_a ($\Delta \approx 17$), suggests that utilizing stronger acids may be potentially detrimental to the polymerization. For example, the reaction catalyzed by perchloric acid displayed the highest monomer conversion (76%) although the isolated polymer possessed significantly lower molecular weight compared to the other acid-catalyzed reactions. We attributed its low molecular weight to degradation and/or undesirable oxidation, as the reaction mixture darkened considerably. Additionally, mineral acids and TFA present safety hazards as they are corrosive, and they may induce heterogeneity in the reaction due to their hygroscopic nature. Consequently, we selected DPP for further experiments due to several advantages. Namely, it is a non-hazardous^[58] organic acid with the highest pK_a (2.04)^[59] of the screened acids. DPP is also an easily weighable solid with a long shelf-life.^[60]

At [EtLp]₀ = 4 M and 5 mol% DPP, we evaluated several solvents to find a greener and safer alternative to DCM. We monitored conversion of the polymerizations using ¹H NMR spectroscopy (Figure 2c). In total, we screened 14 solvents under these conditions, 6 of which are marketed as “green” alternative solvents due to their production from biogenic resources (Figure 2d, Table S2, Supporting Information). Importantly, all solvents were used as received. Solvent choice had a significant impact on both the kinetics and thermodynamics (equilibrium monomer concentration) of the polymerization; however, no

clear relationships to solvent properties appeared (Figure 2d). For example, cyrene and propylene carbonate (PC) showed >50% conversion after 0.5 h but other polar solvents such as dimethyl isosorbide, Rhodiasolve, and DMF were comparatively sluggish (Table S2, Supporting Information). Using DCM, 1,4-dioxane, and acetone also led to relatively fast polymerization kinetics with >40% conversion observed at 0.5 h (Table S2, Supporting Information). Sampling of the polymerizations after 24 h, where equilibrium had been reached, revealed that acetone, PC, and DCM yielded >60% monomer conversion (Figure 2d). Again, the polymerization reactions showed no clear trends among the other solvents (Figure 2d). A deeper mechanistic investigation of solvent effects on polymerization efficiency is ongoing. Nevertheless, we selected acetone to conduct further experiments as it is safer than DCM and possesses both a lower boiling point and viscosity compared to PC, allowing for more convenient polymer isolation and purification. Finally, acetone is more economical than PC as it is one of the highest volume commodity solvents.

2.3. Thermodynamic and Kinetic Parameters of DPP-Catalyzed Polymerization

We assessed thermodynamic characteristics of CROP by measuring the equilibrium monomer concentration under various conditions in acetone and 5 mol% DPP (Figure 3). At 30 °C and with different initial concentrations of EtLp, we monitored conversion by ¹H NMR spectroscopy until reaching equilibrium (24 h) (Figure 3a, Figure S28–31, Table S3, Supporting Information). Decreasing the initial monomer concentration led to lower conversion, which is typical of equilibrium CROP;^[61] this has also been observed in anionic^[38] and radical^[37] ROP of lipoates. Conversion plateaued near 15% at [EtLp]₀ = 1 M; polymer was undetectable at 0.5 M. Interestingly, our results and prior work^[37,38] contradict a recent study on CROP of lipoic acid reporting 80% monomer conversion for polymerizations at 25 °C and [EtLp]₀ = 0.5 M.^[53] Concentration also influenced the molecular weight of the polymer

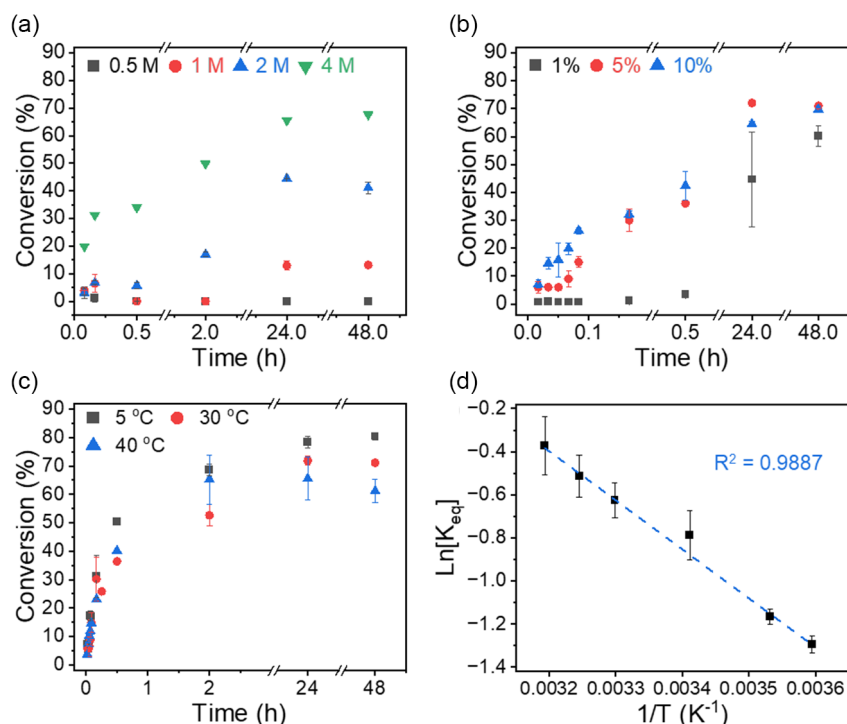


Figure 3. Additional experiments for DPP-catalyzed CROP of EtLp in acetone. a) Plot of conversion versus time for different $[\text{EtLp}]_0$ assayed by ^1H NMR with $n = 3$ samples (Figure S4, S28–S31, Supporting Information). 5 mol% DPP, 30 °C. b) Plot of conversion vs time for different catalyst loading of DPP measured by ^1H NMR with $n = 3$ samples (Figure S4, S34–S39, Supporting Information). $[\text{EtLp}]_0 = 4 \text{ M}$, 30 °C. c) Plot of conversion vs time for different isotherms assayed by ^1H NMR with $n = 3$ samples (Figure S4, S41–S43, Supporting Information). $[\text{EtLp}]_0 = 4 \text{ M}$, 5 mol% DPP. d) Van't Hoff analysis of polymer-monomer equilibrium using variable temperature ^1H NMR with $n \geq 8$ samples (Figure S4, S45, Supporting Information). $[\text{EtLp}]_0 = 4 \text{ M}$, 5 mol% DPP.

leading to a fivefold decrease in M_n as initial monomer concentration decreased from 4 M to 1 M (Figure S32, S33, Table S5, Supporting Information).

We then varied the catalyst loading at benchmark reaction conditions, i.e., $[\text{EtLp}]_0 = 4 \text{ M}$ in acetone at 30 °C (Figure 3b). Unsurprisingly, the rate of conversion showed a positive correlation with mol% DPP; this did not significantly affect the equilibrium conversion (Figure 3b, Figure S34–39, Supporting Information). However, at 1 mol% DPP loading a significant induction period appeared with conversion delayed until ≈ 30 min after the addition of the DPP (Figure S34, Supporting Information). In contrast, little to no induction period occurred at 5 and 10 mol% catalyst loading (Figure S36, S38, Supporting Information). TFA-catalyzed CROP of methyl lipoate under Schlenk conditions^[40] also showed a substantial decrease in conversion and overall efficiency with catalyst loading $\leq 1 \text{ mol\%}$.^[40]

The mol% DPP influences polymer molecular weight, which is unsurprising as DPP likely generates a sulfonium species that then initiates the polymerization.^[40] At 1 mol% DPP, a polymer with $M_n = 700 \pm 200 \text{ kDa}$ forms; at 10 mol% DPP, the molecular weight decreases ($M_n = 300 \pm 30 \text{ kDa}$) (Figure S40, Table S5, Supporting Information). These results corroborate the hypothesis that DPP is also an initiator, generating active propagating species. However, the initiation efficiency is likely poor and variable considering: 1) relatively high-molecular weight is observed for all polymers despite the small ratio of monomer to initiator (only 20:1 to 100:1) and 2) there is a nonlinear relationship between molecular weight and initiator loading.

We considered the possibility of an adventitious initiator species but then discounted that as an analogous polymerization under Schlenk conditions using purified reagents showed no decrease in polymer molecular weight (*vide infra*). We consistently observed similar trends in other polymerizations employing different catalysts (e.g., HCl) and solvents (e.g., DCM). We performed several experiments with different co-initiator species as chain-transfer agents (CTA) to control molecular weight; however, these attempts were largely unsuccessful (Figure S73–78, Table S13, Supporting Information). Our attempt to quantitatively investigate the kinetics at different temperatures failed because of the dynamic nature of the polymerization–depolymerization equilibrium and variable induction periods.

We then performed variable temperature-equilibrium experiments under benchmark conditions. We saw a substantial effect between reaction temperature and polymer molecular weight, with the reaction at 5 °C producing a larger polymer ($M_n = 1400 \pm 500 \text{ kDa}$) than the reaction at 40 °C ($M_n = 430 \pm 80 \text{ kDa}$) (Figure S41–43, Table S5, Supporting Information). However, the kinetics of the polymerization were largely independent of reaction temperature (Figure 3c). We used seven unique temperatures (5–40 °C) (Figure S45, Table S4, Supporting Information) to assess thermodynamic parameters of polymerization (Figure 3d, three values of T shown). The equilibrium constant, K , was derived by measuring relative monomer concentration at each temperature. A Van't Hoff plot, K vs T^{-1} , revealed ΔH° ($-18.8 \pm 1.0 \text{ kJ mol}^{-1}$) and ΔS° ($57 \pm 3 \text{ J mol}^{-1} \text{ K}^{-1}$), respectively (Figure 3d). These values are similar to previous

reports for the ROP of EtLp^[38,40] with the minor deviation likely caused by differences in solvent choice (acetone vs DCM), which impacts monomer–polymer equilibria,^[62] and reaction conditions, i.e., previous studies were under anaerobic conditions. Using these data at $[\text{EtLp}]_0 = 4.0 \text{ M}$, we derived the ceiling temperature ($T_c = 57^\circ\text{C}$); the T_c occurs when equilibrium [monomer] equals initial [monomer]. Importantly, T_c depends on both concentration and solvent, where interactions among solvent, monomer, and polymer contribute to substantially alter the T_c .^[62,63]

The optimal conditions for maximizing conversion and polymer molar mass were observed as $[\text{EtLp}]_0 = 4.0 \text{ M}$ with 5 mol% DPP in acetone at 5°C and 24 h reaction time. As a control, we conducted the polymerization under Schlenk conditions, i.e., a nitrogen atmosphere with dried solvent and reagents. The resulting polymer resembled the polymer from the analogous reaction under ambient atmosphere ($M_n = 1300$ vs 1400 kDa) (Figure S46, Table S5, Supporting Information). However, equilibrium conversion (Figure S47, Supporting Information) was significantly lower than the benchtop reaction ($\approx 49\%$ vs $>60\%$). Similar results were also obtained for a control reaction at 30°C . The reason for this decreased conversion is unclear, but it is consistent with a prior report that also employed Schlenk conditions.^[40] Regardless, using “wet” solvents in an open atmosphere is operationally easier and a more scalable process. Careful control of reaction conditions and reagent purity are not essential to achieve high monomer conversion and produce a UHMW polydisulfide using our approach.

2.4. Closed-Loop Process for Polymerization and Depolymerization

Using optimized conditions, we performed CROP of EtLp on a 50-gram scale to produce polyEtLp for subsequent thermomechanical

analysis and chemical waste analysis (Figure S64, Supporting Information). The E-factor for the polymerization was 9.06, with most of this waste ($\approx 90\%$) coming from diethyl ether used to purify the crude polymer, i.e., remove catalyst and unreacted monomer (Figure S49). The polymer was then hot-pressed (55°C) into thin films suitable for thermomechanical testing (Figure 4a). The polymer underwent minimal degradation after processing as evidenced by the similar molar mass profiles between the pristine and hot-pressed samples (Figure 4b, Figure S50, Supporting Information). Repeated hot-pressing experiments also showed that the polymer could be reprocessed without deterioration as evidenced by minimal depolymerization according to ^1H NMR analyses (Figure S50, Supporting Information). GPC analysis of the reprocessed samples showed similar results; however, there was slight increase in elution time indicating a marginally lower molecular weight (Figure S51, S52, Table S6, Supporting Information). Nevertheless, this robust reprocessing behavior of polyEtLp suggests that the materials may be compatible with extrusion-based recycling methods that are common to industry.

We examined thermomechanical properties of the film using dynamic mechanical analysis (DMA) in tensile mode. UHMW polyEtLp ($M_n = 1300 \text{ kDa}$, $\bar{D} = 1.8$, see Table S4, Supporting Information entry 6) possessed a relatively low-glass transition temperature (T_g), near -32°C according to the maximum of the $\tan \delta$ value (Figure 3c) and around -59°C from differential scanning calorimetry (DSC) (Figure S54, Supporting Information). At temperatures above its T_g , the polymer exhibited a wide rubbery plateau that extended out to $\approx 120^\circ\text{C}$ before the material lost structural integrity; this value was well below the degradation temperature ($T_{d,5\%} = 191^\circ\text{C}$) (Figure S53, Supporting Information). Such behavior is typically observed in rubbery

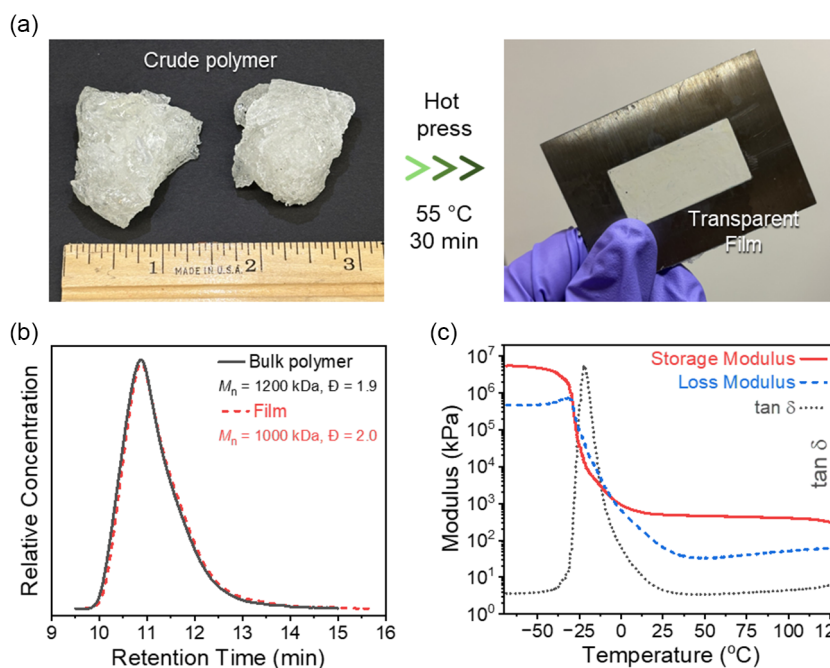


Figure 4. Thermomechanical data for polyEtLp films. a) Reaction scheme for hot pressing polyEtLp into a film. b) GPC chromatograph showing the minimal polymer degradation after processing the bulk polymer sample. c) DMA thermogram of polyEtLp film.

thermosets possessing covalent cross-linking (e.g., natural rubber), thermoplastic elastomers (TPEs),^[64] and semi-crystalline thermoplastics^[65] when the sample is at a temperature above its respective T_g but below the melt temperature (T_m). However, this phenomenon is rarely observed in non-cross-linked, amorphous polymers that are heated well above their T_g . Such polymers typically feature a narrow rubbery plateau and undergo a rapid decrease in viscosity at temperatures slightly greater than their T_g .^[66] Here, the extreme molar mass of the polyEtLp likely leads to extensive chain entanglement^[54] thus strengthening physical cross-linking and giving rise to its distinctive mechanical properties. Finally, these advantageous properties may also be augmented by a polycatenated structure due to a cyclic polymer topology, which was postulated for related polydisulfides.^[67] We are presently studying the topology and morphology of polyEtLp.

We then studied the RCD of polyEtLp with the aim to optimize this depolymerization process by eliminating commonly employed harmful catalysts,^[40] solvents,^[41] and reagents.^[38,68] The depolymerization experiments were initially performed in acetone ([monomer units of polyEtLp]₀ = 0.1 M) at reflux conditions with sub-stoichiometric quantities of various acids such as Amberlyst 15, HCl (12.1 M) or sulfuric acid (18 M) as a catalyst. Although ¹H NMR analysis showed the polymer was efficiently depolymerized with 1 mol% sulfuric acid in acetone to regenerate monomer (Figure S56, Supporting Information), the subsequent isolation and purification of the monomer proved challenging

resulting in significant re-polymerization likely due to residual acid. Switching the solvent to 2-methyl THF or dimethyl isosorbide, which are renewable ethers, resulted in incomplete depolymerization (Figure S57, 58, Supporting Information).

Substituting chloroform as the depolymerization solvent eliminated these issues and monomer was recovered (84% yield, >98% purity) from 5 g of polymer after refluxing for 2 h (Figure S59, S60, and S65, Supporting Information). Polymerization of the regenerated EtLp under the same optimized procedure resulted in a polymer with M_n = 2200 kDa, which is superior to the input material. However, this result was attributed to evaporation of acetone during polymerization, artificially increasing monomer concentration during polymerization. The chemically recycled polymer also showed similar thermomechanical properties comparable to the pristine polymer despite the larger molar mass of the former. Although the RCD process was improved compared to prior studies, future experiments seek to replace chloroform as the solvent.

2.5. LCA of Material Production

We next used LCA^[69–71] to quantify the *cradle-to-gate* environmental impacts of EtLp monomer synthesis from lipoic acid, CROP to yield polyEtLp, and finally the RCD to regenerate monomer (Figure 5a). Our attention first focused on the *climate-change* impact category as the greenhouse gas emissions from

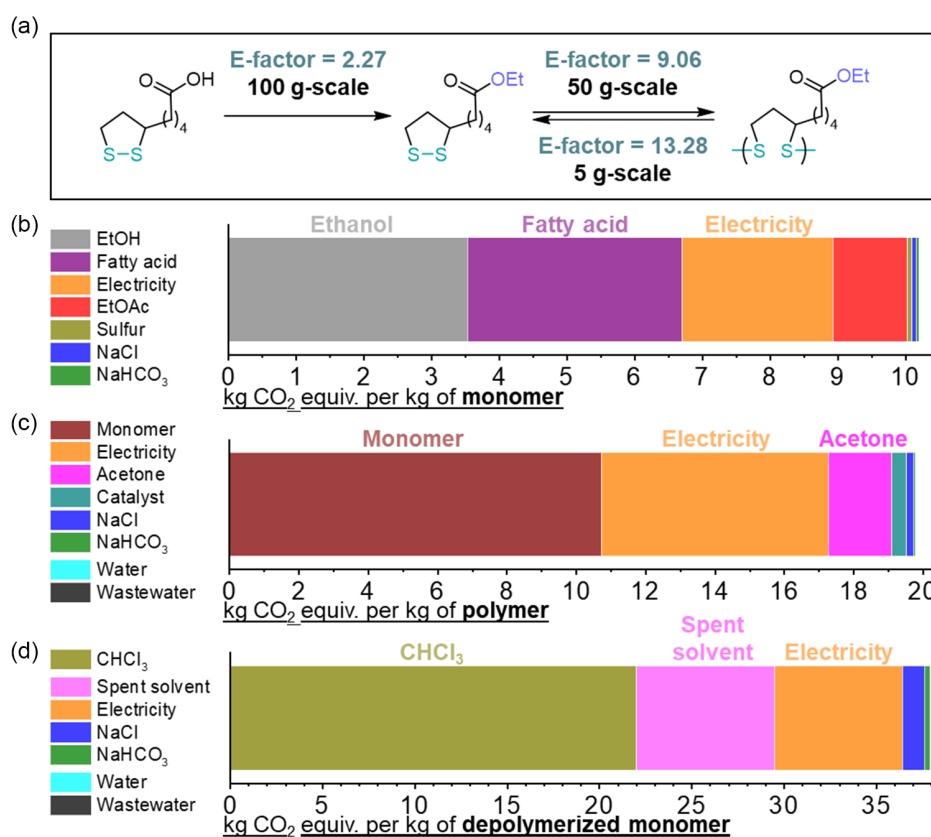


Figure 5. Reaction overview of each process including respective E-factor. Disaggregated *climate change* (quantified in kg CO₂ equiv. per kilo of output material) for b) monomer synthesis, c) polymer synthesis, and d) depolymerization back to monomer.

the life cycle of plastics seriously threaten global climate goals. In fact, under a business-as-usual scenario, the production, disposal, and incineration of plastics were projected at 1.34 gigatons per year by 2030.^[72] Our LCA rendered carbon footprint values of 96.1, 362.3, and 305.0 kg CO₂ equiv kg⁻¹ for EtLp, polyEtLp, and depolymerization back to EtLp, respectively. These values, which are two orders of magnitude higher than CO₂ emissions from conventional plastics, were driven by the energy-intensive nature of laboratory-scale syntheses. In fact, our process required 176, 517, and 548 kWh to obtain a single kilogram of monomer, polymer, and recovered monomer after depolymerization, respectively. We repeated the calculations by replacing the medium voltage for the U.S. Grid Region in Virginia with a model 100% renewable energy grid and obtained disaggregated values of *climate-change* potential (Table S7–S9, Supporting Information); this moderates some of the inflated impact of inefficient energy consumption from operating the processes at the laboratory scale.

The EtLp synthesis from a Fischer esterification of lipoic acid rendered a carbon footprint of 10.2 kg CO₂ equiv kg⁻¹ (Figure 5b). Ethanol (36.6%), fatty acid (31.1%), and electricity consumption (21.8%) represented the bulk of emissions (note that lipoic acid was modeled as a combination of fatty acid and sulfur (Figure S63, Table S7, Supporting Information). Such results originated from the extensive use of these chemicals and energy costs, where ≈1.6 kg of ethanol and ≈0.8 kg of fatty acid are required to yield a single kilogram of monomer. The polymer production showed a *climate-change* potential of 19.8 kg CO₂ equiv kg⁻¹ with significant contributions from monomer synthesis (54.2%), electricity consumption (33.0%), and acetone (9.3%) (Figure 5c). Ang et al. obtained an even higher value ranging from 360 to 1750 kg CO₂ equiv kg⁻¹ when considering the laboratory-scale synthesis of one kilogram of poly(ε-caprolactone) using various methods, including CROP.^[73] Data for these processes show polyEtLp, and related derivatives are promising low-footprint materials.

The cationic RCD of polyEtLp emitted 38.0 kg CO₂ equiv kg⁻¹ to regenerate EtLp monomer (Figure 5d). This result seemed

counterintuitive because the depolymerization back to monomer generated a fourfold increase in CO₂ emissions compared to monomer synthesis from virgin resources. However, a prior study reported similar observations for the upcycling of poly(ethylene terephthalate) (PET) via chemo-selective depolymerization; the *climate change* values ranged from 4.3 to 91.3 kg CO₂ equiv kg⁻¹ compared to only 3.0 kg CO₂ equiv kg⁻¹ for pristine PET production.^[74] Furthermore, virgin terephthalic acid (monomer) and PET production outperformed enzymatic hydrolysis of PET (1.2–17 fold worse), and most emissions were primarily due to the extensive use of NaOH and electricity.^[75] In our case, the greater carbon footprint for polyEtLp RCD resulted from the use (58% of emissions) and post-treatment (20% of emissions) of the hazardous solvent chloroform. Continued optimization of the depolymerization reaction (e.g., replacing chloroform) should significantly lower impacts of this process.

To gain broader perspective on the environmental impacts of polyEtLp synthesis and its closed-loop recycling, we assessed other relevant impact categories using TRACI 2.1^[76] and cumulative energy demand (CED) methodologies against several commodity polymers (Figure 6, Figure S66, Tables S10, S11, Supporting Information). The *climate-change* impact of polyEtLp was within one order of magnitude of those for conventional petrol-based plastics and comparable to that of polydimethylsiloxane, which is a commodity elastomer. Impact categories of *eutrophication*, *photochemical oxidant formation*, and *particulate matter formation* provided a similar picture, where polyEtLp scored slightly above (though within one order of magnitude of) the commercial materials. Since the laboratory-scale production of polyEtLp yielded comparatively higher CED, the impact of polyEtLp may be dramatically reduced by implementing optimized industrial-scale processes. Interestingly, the RCD of polyEtLp to monomer also showed reduced impacts (compared to synthesized polymer) in three of the six impact categories that we analyzed (Figure S66, Table S10, Supporting Information). Thus, closed-loop recycling may further reduce impacts in categories other than CO₂ emissions. It is important to mention that the RCD of polyEtLp was assessed at a relatively small scale

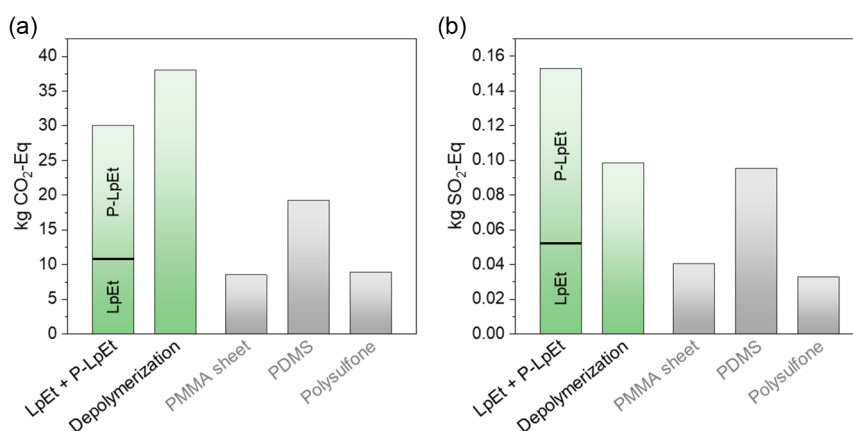


Figure 6. Environmental impacts in the categories of a) *climate change* and b) *acidification* of monomer synthesis, polymer synthesis, and depolymerization back to monomer (indicated in green) according to TRACI 2.1 impact assessment.^[76] The results corresponding to benchmark commercial polymers are indicated in gray. Additional metrics are shown in the supporting information (Figure S66, Supporting Information).

(5 gram-scale) due to the large volume of solvent required in our batch reaction method. However, switching from a batch reaction to a flow reactor process could mitigate this apparent limitation on scalability.

Considering data from the LCA analysis, we finally explored several proof-of-concept experiments to obtain additional efficiency gains. In the first experiment, we conducted a one-pot polymer synthesis from lipoic acid. The residual sulfuric acid used in the Fischer esterification served as the polymerization catalyst resulting in a UHMW polymer ($M_n = 1100$ kDa) with moderate dispersity (Figure S67, Table S12, Supporting Information). We then performed a polymerization under reduced pressure at $\approx 5^\circ\text{C}$ on a rotary evaporator where acetone was removed *in situ*. The resulting polymer possessed a larger molecular weight ($M_n = 1700$ kDa, $\bar{D} = 1.7$) compared to materials obtained in previous methods (Figure S69, Table S12, Supporting Information). In the final experiment, we conducted an RCD of polyEtLp in chloroform and subsequently distilled the solvent to induce CROP *in situ*. The regenerated pristine polymer ($M_n = 600$ kDa) was produced with high conversion (80%), thus avoiding waste associated with isolation and purification of the monomer (Figure S71, Table S12, Supporting Information). In all cases, we obtained high-molecular-weight polymers, which were analogous to polymers obtained under standard solution-phase polymerizations. These experiments provide preliminary proof-of-concept data that the processes can be further optimized in an industrial setting.

3. Conclusion

In summary, the disclosed methodology for the scalable Fischer esterification of lipoic acid and the subsequent DPP-catalyzed CROP outlines a more industrially viable process route compared to existing methods. It also closely adheres to the guiding principles of Green Chemistry, which are often overlooked in both monomer and polymer production. LCA indicates that our laboratory-scale process approaches the environmental footprint of many commercial polymers. Importantly, these processes can be further optimized, as shown in the final proof-of-concept experiments. We anticipate added efficiency gains if the processes are conducted at even larger scale. Additional studies to elucidate solvent effects on polymerization efficiency may also afford more optimized design processes. Despite these advantages, we need additional work in 1) characterizing morphological properties of polyEtLp itself and 2) expanding the scope of monomers to access polydisulfides with tunable bulk properties. Recent work suggests that high-melting-point alcohols such as isosorbide or menthol^[41] may be compatible with Fischer esterification protocols, potentially providing additional monomers at useful quantities, although these processes remain to be optimized or scaled up. Regardless, the circular nature of 1,2-dithiolanes and their corresponding polydisulfide materials have high potential as more sustainable polymers, especially as processes for their production and disposal are increasingly scaled.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Joshua C. Worch conceived the work and directed the research. Joshua C. Worch and Christopher B. Koelbl designed the experiments. Christopher B. Koelbl, Erlantz Lizundia, Chizoba Obunadike, Woojung Ham, Nadim Mahmud, and Mathew Garcia performed and analyzed experiments. Joshua C. Worch, Christopher B. Koelbl, and Erlantz Lizundia prepared the manuscript, and all authors contributed to manuscript revisions.

Data Availability Statement

All data are available in the manuscript or supplementary information.

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- [1] F. Vidal, E. R. van der Marel, R. W. F. Kerr, C. McElroy, N. Schroeder, C. Mitchell, G. Rosetto, T. T. D. Chen, R. M. Bailey, C. Hepburn, C. Redgwell, C. K. Williams, *Nature* **2024**, *626*, 45.
- [2] J.-G. Rosenboom, R. Langer, G. Traverso, *Nat. Rev. Mater.* **2022**, *7*, 117.
- [3] I. R. Campbell, M.-Y. Lin, H. Iyer, M. Parker, J. L. Fredricks, K. Liao, A. M. Jimenez, P. Grandgeorge, E. Roumeli, *Annu. Rev. Mater. Res.* **2023**, *53*, 81.
- [4] Z. Wang, M. S. Ganewatta, C. Tang, *Prog. Polym. Sci.* **2020**, *101*, 101197.
- [5] Y. Zhu, C. Romain, C. K. Williams, *Nature* **2016**, *540*, 354.
- [6] J. C. Worch, A. P. Dove, *ACS Macro Lett.* **2020**, *9*, 1494.
- [7] A. Rahimi, J. M. Garcia, *Nat. Rev. Chem.* **2017**, *1*, 0046.
- [8] S. C. Kosloski-Oh, Z. A. Wood, Y. Manjarrez, *Mater. Horiz.* **2021**, *8*, 1084.
- [9] C. Jehanno, J. W. Alty, M. Roosen, S. De Meester, A. P. Dove, E. Y. X. Chen, F. A. Leibfarth, H. Sardon, *Nature* **2022**, *603*, 803.
- [10] L. T. J. Korley, T. H. Epps, B. A. Helms, A. J. Ryan, *Science* **2021**, *373*, 66.
- [11] A. Dev, A. Rösler, H. Schlaad, *Polym. Chem.* **2021**, *12*, 3084.
- [12] K. Ratzenböck, M. M. Ud Din, S. M. Fischer, E. Žagar, D. Pahovnik, A. D. Boese, D. Rettenwander, C. Slugovc, *Chem. Sci.* **2022**, *13*, 6920.
- [13] T. M. Bennett, J. Portal, V. Jeanne-Rose, S. Taupin, A. Ilchev, D. J. Irvine, S. M. Howdle, *Eur. Polym. J.* **2021**, *157*, 110621.
- [14] T. Erdmenger, C. Guerrero-Sanchez, J. Vitz, R. Hoogenboom, U. S. Schubert, *Chem. Soc. Rev.* **2010**, *39*, 3317.
- [15] A. Basterretxea, E. Gabirondo, C. Jehanno, H. Zhu, O. Coulembier, D. Mecerreyes, H. Sardon, *Macromolecules* **2021**, *54*, 6214.
- [16] J. C. Worch, A. P. Dove, *Acc. Chem. Res.* **2022**, *55*, 2355.
- [17] Z. Geng, J. J. Shin, Y. Xi, C. J. Hawker, *J. Polym. Sci.* **2021**, *59*, 963.
- [18] P. Anastas, N. Eghbali, *Chem. Soc. Rev.* **2010**, *39*, 301.
- [19] P. T. Anastas, J. C. Warner, *Green chemistry: theory and practice*, Oxford University Press, Oxford, **2000**.
- [20] D. E. Fagnani, J. L. Tami, G. Copley, M. N. Clemons, Y. D. Y. L. Getzler, A. J. McNeil, *ACS Macro Lett.* **2021**, *10*, 41.

- [21] J. Wolfs, M. A. R. Meier, *Green Chem.* **2021**, *23*, 4410.
- [22] J. Demarteau, B. Cousineau, Z. Wang, B. Bose, S. Cheong, G. Lan, N. R. Baral, S. J. Teat, C. D. Scown, J. D. Keasling, B. A. Helms, *Nat. Sustain.* **2023**, *6*, 1426.
- [23] L. P. Manker, G. R. Dick, A. Demongeot, M. A. Hedou, C. Rayroud, T. Rambert, M. J. Jones, I. Sulaeva, M. Vieli, Y. Leterrier, A. Potthast, F. Maréchal, V. Michaud, H.-A. Klok, J. S. Luterbacher, *Nat. Chem.* **2022**, *14*, 976.
- [24] C. Libretti, M. A. R. Meier, *Macromolecules* **2023**, *56*, 7532.
- [25] G. W. Coates, Y. D. Y. L. Getzler, *Nat. Rev. Mater.* **2020**, *5*, 501.
- [26] K. A. Stellmach, M. K. Paul, M. Xu, Y.-L. Su, L. Fu, A. R. Toland, H. Tran, L. Chen, R. Ramprasad, W. R. Gutekunst, *ACS Macro Lett.* **2022**, *11*, 895.
- [27] C. Hardy, G. Kociok-Köhn, A. Buchard, *Polym. Chem.* **2023**, *14*, 623.
- [28] Z. Hu, S. N. Bernsten, C. Shi, A. Sangroniz, E. Y. X. Chen, G. M. Miyake, *Chem* **2024**, *10*, 3040.
- [29] L. Zhou, Z. Zhang, C. Shi, M. Scoti, D. K. Barange, R. R. Gowda, E. Y. X. Chen, *Science* **2023**, *380*, 64.
- [30] W. J. Neary, J. G. Kennemur, *Macromolecules* **2017**, *50*, 4935.
- [31] D. Sathe, H. Chen, J. Wang, *Macromol. Rapid Commun.* **2023**, *44*, 2200304.
- [32] G. Rosetto, F. Vidal, T. M. McGuire, R. W. F. Kerr, C. K. Williams, *J. Am. Chem. Soc.* **2024**, *146*, 8381.
- [33] B. A. Abel, R. L. Snyder, G. W. Coates, *Science* **2021**, *373*, 783.
- [34] J. A. Barltrop, P. M. Hayes, M. Calvin, *J. Am. Chem. Soc.* **1954**, *76*, 4348.
- [35] Q. Zhang, D.-H. Qu, B. L. Feringa, H. Tian, *J. Am. Chem. Soc.* **2022**, *144*, 2022.
- [36] M. Pięta, V. B. Purohit, J. Pietrasik, C. M. Plummer, *Polym. Chem.* **2023**, *14*, 7.
- [37] M. Raeisi, N. V. Tsarevsky, *J. Polym. Sci.* **2021**, *59*, 675.
- [38] Y. Liu, Y. Jia, Q. Wu, J. S. Moore, *J. Am. Chem. Soc.* **2019**, *141*, 17075.
- [39] T. Du, B. Shen, J. Dai, M. Zhang, X. Chen, P. Yu, Y. Liu, *J. Am. Chem. Soc.* **2023**, *145*, 27788.
- [40] B. S. Wang, Q. Zhang, Z. Q. Wang, C. Y. Shi, X. Q. Gong, H. Tian, D. H. Qu, *Angew. Chem., Int. Ed.* **2023**, *62*, e202215329.
- [41] T. O. Machado, C. J. Stubbs, V. Chiaradia, M. A. Alraddadi, A. Brandolese, J. C. Worch, A. P. Dove, *Nature* **2024**, *629*, 1069.
- [42] M. A. Alraddadi, V. Chiaradia, C. J. Stubbs, J. C. Worch, A. P. Dove, *Polym. Chem.* **2021**, *12*, 5796.
- [43] Q. Zhang, Y. Deng, C.-Y. Shi, B. L. Feringa, H. Tian, D.-H. Qu, *Matter* **2021**, *4*, 1352.
- [44] Q. Zhang, C.-Y. Shi, D.-H. Qu, Y.-T. Long, B. L. Feringa, H. Tian, *Sci. Adv.* **2018**, *4*, eaat8192.
- [45] I. O. Levkovsky, L. Trachsel, H. Murata, K. Matyjaszewski, *ACS Macro Lett.* **2025**, *14*, 207.
- [46] Y. Okayama, P. Morris, K. Albanese, S. Olsen, A. Mori, J. R. de Alaniz, C. M. Bates, C. J. Hawker, *J. Polym. Sci.* **2025**, *63*, 1345.
- [47] K. R. Albanese, P. T. Morris, J. Read de Alaniz, C. M. Bates, C. J. Hawker, *J. Am. Chem. Soc.* **2023**, *145*, 22728.
- [48] K. R. Albanese, Y. Okayama, P. T. Morris, M. Gerst, R. Gupta, J. C. Speros, C. J. Hawker, C. Choi, J. R. de Alaniz, C. M. Bates, *ACS Macro Lett.* **2023**, *12*, 787.
- [49] H. Tang, N. V. Tsarevsky, *Polym. Chem.* **2015**, *6*, 6936.
- [50] D. Woo, N. Kang, H. Park, S. Myoung, G. Lee, J. Kon Kim, C. Choi, *Chem. Eng. J. Lausanne* **2024**, *488*, 150818.
- [51] L. Field, R. B. Barbee, *J. Org. Chem.* **1969**, *34*, 36.
- [52] F. S. Dainton, J. A. Davies, P. P. Manning, S. A. Zahir, *Trans. Faraday Soc.* **1957**, *53*, 813.
- [53] H.-T. Zhang, L.-K. Hou, G.-W. Chu, J.-X. Wang, L.-L. Zhang, J.-F. Chen, *Chem. Eng. J. Lausanne* **2024**, *482*, 148816.
- [54] D. C. Kong, M. H. Yang, X. S. Zhang, Z. C. Du, Q. Fu, X. Q. Gao, J. W. Gong, *Macromol. Mater. Eng.* **2021**, *306*, 2100536.
- [55] A. Jordan, K. D. Whymark, J. Sydenham, H. F. Sneddon, *Green Chem.* **2021**, *23*, 6405.
- [56] EPA, *Final Risk Evaluation for Methylene Chloride*, United States Environmental Protection Agency, **2024**, <https://www.epa.gov/assessing-and-managing-chemicals-under-tsca/final-risk-evaluation-methylene-chloride>.
- [57] R. A. Sheldon, *Green Chem.* **2007**, *9*, 1273.
- [58] Diphenyl phosphate, No. 850608 Sigma-Aldrich, **2024**, <https://www.sigmaaldrich.com/US/en/sds/aldrich/850608>, (accessed August 2024).
- [59] M. A. F. Delgove, A. A. Wróblewska, J. Stouten, C. A. M. R. van Slagmaat, J. Noordijk, S. M. A. De Wildeman, K. V. Bernaerts, *Polym. Chem.* **2020**, *11*, 3573.
- [60] K. Makiguchi, T. Satoh, T. Kakuchi, *Macromolecules* **2011**, *44*, 1999.
- [61] P. Olsén, K. Odelius, A.-C. Albertsson, *Biomacromolecules* **2016**, *17*, 699.
- [62] L. Cederholm, J. Wohler, P. Olsén, M. Hakkarainen, K. Odelius, *Angew. Chem., Int. Ed.* **2022**, *61*, e202204531.
- [63] P. Olsén, J. Undin, K. Odelius, H. Keul, A.-C. Albertsson, *Biomacromolecules* **2016**, *17*, 3995.
- [64] G. Holden, in *Applied Plastics Engineering Handbook*, 3rd ed., (ed: M. Kutz), William Andrew Publishing, Oxford, **2024**, pp. 97, <https://doi.org/10.1016/B978-0-323-88667-3.00020-5>.
- [65] E. Lizundia, S. Petisco, J.-R. Sarasua, *J. Mech. Behav. Biomed. Mater.* **2013**, *17*, 242.
- [66] S. J. Stadnicki, J. K. Gillham, R. F. Boyer, *J. Appl. Polym. Sci.* **1976**, *20*, 1245.
- [67] H. Ishida, A. Kisanuki, K. Endo, *Polym. J.* **2009**, *41*, 110.
- [68] M. M. Kristensen, K. B. Løvschall, A. N. Zelikin, *ACS Macro Lett.* **2023**, *12*, 955.
- [69] A. Laurent, B. P. Weidema, J. Bare, X. Liao, D. Maia de Souza, M. Pizzol, S. Sala, H. Schreiber, N. Thonemann, F. Veronesi, *J. Ind. Ecol.* **2020**, *24*, 986.
- [70] J. Porzio, C. D. Scown, *Adv. Energy Mater.* **2021**, *11*, 2100771.
- [71] J. F. Peters, *Nat. Sustain.* **2023**, *6*, 614.
- [72] C. Law, *Plastic and Climate: The Hidden Costs of A Plastic Planet.*, Center for International and Environmental Law **2019**, <https://www.ciel.org/project-update/plastic-climate-the-hidden-costs-of-a-plastic-planet/>.
- [73] P. Ang, S. R. Mothe, L. R. Chennamaneni, F. Aidil, H. H. Khoo, P. Thoniyot, *ACS Sustainable Chem. Eng.* **2021**, *9*, 669.
- [74] M. Iturrondobeitia, L. Alonso, E. Lizundia, *Resour. Conserv. Recycl.* **2023**, *198*, 107182.
- [75] T. Uekert, J. S. DesVeaux, A. Singh, S. R. Nicholson, P. Lamers, T. Ghosh, J. E. McGeehan, A. C. Carpenter, G. T. Beckham, *Green Chem.* **2022**, *24*, 6531.
- [76] J. Bare, *Clean Technol. Environ. Policy* **2011**, *13*, 687.

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